NIGAH
AI VOICE MOBILITY ASSISTANT

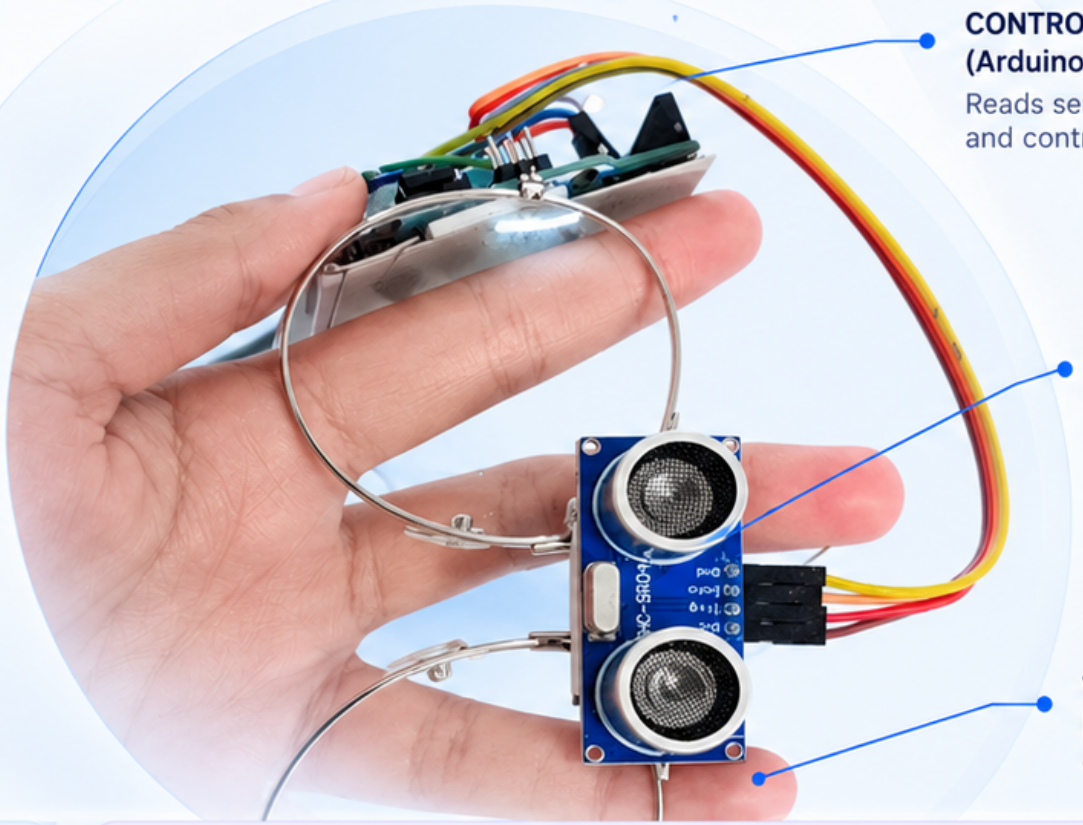
ONE PLATFORM. **MULTIPLE REAL-WORLD APPLICATIONS.**

NIGAH

AI VOICE MOBILITY ASSISTANT

ONE CORE TECHNOLOGY.
MULTIPLE REAL-WORLD APPLICATIONS.

NIGAH is a proposed wearable assistive-technology platform that combines camera input, sensors, AI/decision logic and audio feedback to support different everyday needs.



CONTROLLER (Arduino / ESP32)
Reads sensor data and controls the system.

ULTRASONIC SENSORS
Obstacle detection and distance sensing.

WEARABLE FORM
Lightweight and comfortable.

NIGAH VISION
Obstacle awareness and selected object identification.

NIGAH CARE
Memory assistance, saved item locations and daily reminders.

CAMERA + SENSORS
Capture environment

→

AI / DECISION LOGIC
Understand & prioritize

→

VOICE GUIDANCE
Provide actionable alerts

NIGAH
AI VOICE MOBILITY ASSISTANT

USE CASES & TARGET CUSTOMERS

APPLICATION	USE CASE	TARGET CUSTOMERS
NIGAH VISION	Obstacle alerts, selected object identification	Blind/low-vision people, senior citizens with poor eyesight
NIGAH CARE	"Phone kahan rakha?", saved item locations, medicine and daily-task reminders	People living with dementia, people with memory difficulties, senior citizens
ELDERLY ASSISTANCE	Nearby obstacle alerts and reminders	Families, home-care services, old-age homes
HANDS-FREE ASSISTANCE	Identifying selected tools or objects	Technicians, electricians, maintenance workers
WORKPLACE SAFETY	Nearby obstacle or equipment alerts	Factories, warehouses, construction companies
NAVIGATION ASSISTANCE	Assistance in unfamiliar indoor environments	Hospitals, large campuses, rehabilitation centres

ONE PLATFORM. MULTIPLE REAL-WORLD APPLICATIONS.

COMMERCIAL CUSTOMER SEGMENTS

B2C
Individuals and their families.

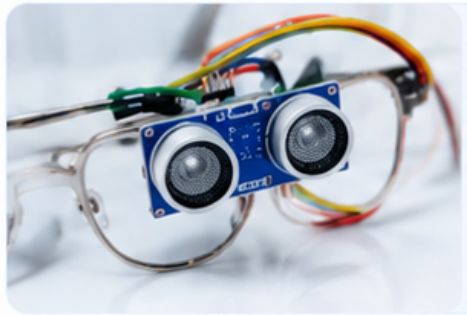
B2B
Hospitals, elder-care facilities, factories and warehouses.

B2B2C / INSTITUTIONAL
NGOs, rehabilitation centres and disability-support organisations.

Note: These are potential use cases and customer groups to investigate, not yet validated customers or proven demand.

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HOW THE SAME CORE TECHNOLOGY CAN BE REUSED



CAMERA + DISTANCE SENSORS
Capture visual information and nearby obstacles.

→

AI / DECISION LOGIC
Interpret sensor input or process a saved reminder.

→

VOICE / AUDIO FEEDBACK
Provide alerts, prompts and reminder playback.

EXAMPLES OF REUSING THE CORE SYSTEM



Camera identifies a supported object

→ **Vision assistance**



Previously saved item location is retrieved

→ **NIGAH Care**



Sensor detects a nearby obstacle

→ **Elderly assistance**



Equipment or nearby obstacles are detected

→ **Workplace safety**



Scheduled reminder is triggered

→ **Medicine or daily-task reminders**

IMPORTANT TECHNICAL LIMITATION

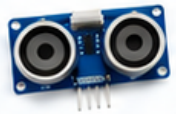
Every use case will not work with exactly the same hardware and software. Some may need extra sensors, a different AI model, GPS, more storage or additional testing. GPS and a smartphone app are possible future extensions, not mandatory for every use case.

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COMPONENTS, REUSE & MINIMUM BUDGET

Proposed hardware for the next prototype

PROPOSED MINIMUM HARDWARE

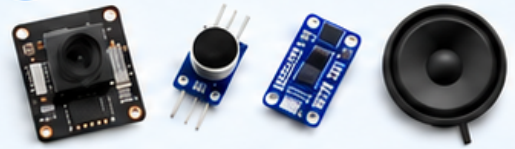
COMPONENT	IMAGE	PURPOSE	USE
ESP32-S3 N16R8 with integrated OV3660 camera		Camera input and constrained edge processing	Vision; foundation for future Care features
2 x HC-SR04 ultrasonic sensors (already owned)		Distance measurement	Obstacle awareness
MAX98357A amplifier + small speaker		Audio output	Alerts and reminders
I2S microphone (e.g. INMP441)		Voice input	Voice/basic voice commands
Battery + charging & regulation module		Power the wearable electronics	All applications
Resistors / voltage divider, wires, connectors		Safe sensor interfacing	Hardware integration
Optional RTC module		Keep scheduled reminders	Care
Optional microSD module/card		Additional local storage for audio/data	Care (if required)

ALREADY AVAILABLE



- Arduino UNO
- Two HC-SR04 sensors
- Breadboard, jumper wires
- Basic components

PROPOSED NEW PURCHASES



- ESP32-S3 camera board
- Microphone
- Audio amplifier + speaker
- Suitable power components

ESTIMATED ADDITIONAL HARDWARE BUDGET

₹2,500 – ₹3,500+

Preliminary estimate for the basic build. Actual prices and compatibility must be checked before purchasing.

FEASIBILITY WITH THE PROPOSED COMPONENTS

Obstacle alerts	Suitable for an initial prototype using HC-SR04 sensors and audio output.
Selected object identification	Requires testing the exact camera board and a lightweight model; real-time performance is not guaranteed.
Saved item locations (NIGAH Care)	Basic voice-note or manual location tagging may be possible, but searchable offline voice memory needs additional software, storage and testing.
Medicine reminders	Scheduled audio reminders are feasible with a suitable RTC and storage. Medicine intake must be confirmed by the user.
Workplace safety	A future exploratory application requiring application-specific sensors, logic and safety validation.

DESIGN CONSTRAINT

The HC-SR04 echo output is typically 5 V, while ESP32-S3 GPIO uses 3.3 V logic. A suitable voltage divider or level-shifting circuit is required.
The Arduino UNO can help with initial sensor experiments, but it may not be necessary in the final wearable setup.

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CASE STUDY, MARKET CONTEXT & NEXT STEPS

Learning from existing solutions and exploring future opportunities

EXISTING SOLUTIONS AND RESEARCH



Envision Glasses

An existing assistive-glasses product to study for visual assistance, product design and user interaction.

Product information ↗

Visual assistance



OrCam MyEye

Another reference for camera-based assistance and audio feedback.

Product information ↗

Object identification & reading



Dementia and memory assistance research

Review reminder technologies, safety monitoring and wandering-related research to understand existing approaches, limitations and user needs.

Research review ↗ 2026 review ↗

Memory assistance & reminders

COMMERCIAL CUSTOMER SEGMENTS

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WHAT COULD MAKE NIGAH WORTH EXPLORING?

Modular approach
One core platform adapted to different user needs.

Affordable solution
Focus on affordability and local user requirements.

Staged development
Validate the Vision prototype first, then evaluate Care and workplace applications.

User-centric design
Component-cost analysis and user feedback before commercial claims.

NEXT STEPS

01 Verify component compatibility and actual purchase costs.

02 Build and test the basic obstacle-alert prototype.

03 Evaluate camera-based object identification on the ESP32-S3.

04 Explore saved-location and reminder functions for NIGAH Care.

05 Conduct user interviews with potential users, families and relevant organisations.

Note: *These are potential use cases and customer groups to investigate, not yet validated customers or proven demand.*